

Evaluation Techniques

- 1) Identify the problem
- 2) design the solution to address the problem
- 3) verify the solⁿ / Evaluation
 - functional / task / feature's accuracy

a) Need to ensure before release → security
 (Is your solⁿ secure enough)

Imp terms

Usability, accessibility, user experience, UX

→ user
 can access
 if possible
 (including all)
 (port. msg should
 be passed to all)

emotion, more
 than usability

'Must ensure'
 things to
 build a SW

* Functionality, reliability, security, usability & UX.

• web@com → prev. versions of applications like Youtube
 (major change in v1 with respect to UX).

• Evaluation → tests usability & functionality
 → evaluates both design & implementation
 → all stage → consider all things

• goals of evaluation

Review-based evaluation

Evaluating designs

① Cognitive walk-through : 1992 ② introduce

• Task-centered design

- per task - module design (prev days)

now task is more dependent on user specification.

• Is a process to evaluate design on how well users can understand / interact / learn the system.

• psychological principles / cognitive psychology → for high level s/w's.

• expert evaluation

• forms use error log to evaluate.

• imagines user's thoughts / actions when they use interface for the 1st time.

• user knowledge in perspective of interaction

from error story design error log
(believable story).

3 steps :

1) select task → depends on time, budget, expertise.

2) write down the desired path

- know the purpose + users + functionality.

3) 4 ques.

[1. begins with ...]

robot

Status msg

case :

Real time video & upload in portfolio

1. Record video

- a1) yes
- a2) yes
- a3) yes
- a4) No

record button
after click

show some timer/
blinking sign → design may
provide some symbol/
indication of
this control.

3. Upload video

2. Stop

- a1) Yes
- a2) yes
- a3) No

a4) No → status
msg add

→ stop button

designer can add **stop** button.

Ask the 4 ques for each story :

yes → No need to take action

No →

Q1 customer click after click → goal set after click

Q2 control visible after click

Q3 control & action → strong link

Q4 will user understand the feedback.



Idea :

recording ongoing

↓ stop

recording stopped !

- 1) Task ↑ cog x hen ✓
 2) blind cog ✓ " x
 3) severity x cog ✓ hen x

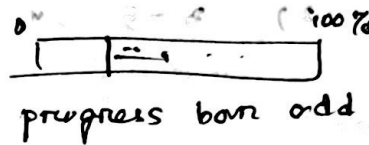
3. Upload video

a1) Yes (obv. video upload button)

a2) No (control visible to user)
 For uploading, we can add an icon under the stop button

a3) Yes (It follows the design recommendation in 3(a2))

a4) No (icon not clearly visible)



→ upload complete (status msg)

Q. sketch the application from given story & show the cognitive walkthrough using 3 vls.

cognitive walkthrough:

- no need for users

user's task → explicitly used

- suggestions ✓

- maintain stage to use possible

- quick & inexpensive.

disadv: - 3-5 experts → optimal result
 >5 → money ↑, right solⁿ suggestion ↑

- Focus on words / graphics

- problem → repeated or, impact huge { freq, persistency, impact
 severity depend on it

priority define karna hai na
 - artificial context

Heuristic Evaluation :

expertise dependent

- highly adopted method

- 10 heuristics

- developed by Nielson

- multiple evaluators → why?

1) 1 evaluator → cannot identify

all problems.

2) 3-5 →

can identify all problems as evaluators are many.

↓
aggregated after collecting problems.

Process - 5 steps :

1. System review, user

2. heuristic set select

3. evaluate

4. communicate & aggregate things / findings.

5. Problems → categorize, prioritize findings.

Severity Rating of HE :

Problem $\rightarrow$ freq. $\rightarrow$ how often occur
 $\uparrow$ $\rightarrow$ high, severe

Impact $\rightarrow$ severe

minor, severity $\downarrow$

persistence $\rightarrow$ repeatedly problem occur

0-4 rating

$\rightarrow$ 0 : Not a usability problem.

$\rightarrow$ 1 : cosmetic problem $\rightarrow$ extra time to fix

$\rightarrow$ 2 : Minor usability problem $\rightarrow$ fix is low priority.

$\rightarrow$ 3 : Major " $\rightarrow$ important to fix

$\rightarrow$ 4 : Usability catastrophe $\rightarrow$ release or you need to fix.

- ⑧ Problems → multiple violation of heuristics.
- design recommendation to address problem
 - severity rating?
 - serial no, location of problem, problem, evidence, violation heuristic, soln, concern.

Heuristic :

- adv →
1. Inexpensive
 2. Expert skills - not a must.
 3. Only stage can be used.
- interface more efficient in later.

- disadv →
1. evaluator → multiple - better.
 2. Not as effective as usability testing.
(assumption of experts)
 3. always ready suggestion (as customer involved).

4. False alarm

10 heuristics :

1. visibility of system status :
 - interface should be self-explainable
2. match betwⁿ system & real world.
 - real world convention follow krni.



Recycle bin

- fees/cost nln 2 rkn.
- No need to use button.
- IUPUI → full forum ?

3. User control & Freedom

undo/redo, etc. option (Gmail)

you can exit the system whenever you want.
comment → feedback → a set of options, restrictive options.



not time to type the

4.

the system, etc. doing that

most of top <> on the



area for the system to

the system to

the system to

the system to

the system to

the system to

the system to

the system to

the system to

4. Consistency & Standards

* External → BD → all classroom

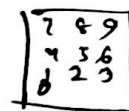


* Internal → all dept of mist has same sitting arrangement.

- use icons that people are familiar with,
- About us >> get to know the team.

↓
internet domain

- respect sentiment of users.



calculator

5. Error Prevention :

- through design, add status messages to prevent error.

6. Recognition rather than recall :

- memory load ↓

design → helps → recognise system.

↓
by using familiar icons, option actions.

- use mouse-over-tooltips.

7. Flexibility & efficiency of use

- one level of efficiency $\rightarrow$ $\uparrow$ usability.
- But there is a tradeoff betwⁿ efficiency & usability $\rightarrow$ Flexibility

Ex: Voice recognition,

accelerators $\rightarrow$ for expert

shortcuts $\rightarrow$ ctrl + c $\rightarrow$ copy

ctrl + x $\rightarrow$ cut

ctrl + v $\rightarrow$ paste

8. Aesthetic & minimalist design

- aesthetic $\uparrow$ user trust, usability $\uparrow$
- but, we have to maintain the main purpose.

eg - minimalist design $\rightarrow$ google search page.

image

Search here...

- 9. Help users recognise, diagnose & recover from errors.

invalid email $\rightarrow$ error msg $\rightarrow$ @...com

10. Help & documentation

- helps to understand the interface

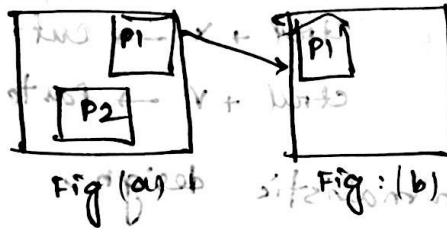
Ex: in-line help

A way for doing a heuristic evaluation:

① ss's of UI → heuristic evaluation
(define problems & evaluate)
or → Are H1, H3, H5, H10 sufficient to assess these UIs? If not, justify.

- ① Problem no. → ② Where → ③ Problem? → ④ Evidence
↓
heuristics violate (Ex: H2, H7)
- ⑤ severity rating (1-4) → ⑥ design recommendation → ⑦ relation/related with → ⑧ remarks (optional)

Excel → 8 columns → Problems are noted.



Ex: Problem 1 (P1) — Fig (a), (b) — system starts — H1
cannot be properly understood.

related Problems

to help with categorization

P2
P3
P4

— P1 —

3/5 person did this table individually

— aggregate problem of E1, E2 ... En

— identify unique problems by reviewing discussing with the team.

→ create report from this

Ans for Q: ~~What~~ heuristics needed to assess UIs.

Q: why is H5 needed in a practical scenario?
Evaluate with one example

Ans: H5 is a heuristic that helps in evaluating the usability of a product.

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Evaluating through user participation

Laboratory → controlled way → lab machine (person researches)
 Field studies → metro rail → (ticket counter)

Laboratory studies : adv, dis

Exposure variable (control content) } bad for feedback.
 confirmant " fix ३१२३ २३३ ३१

comparison should only be done betwⁿ same environment
 setup २ २३३३ things.

Field studies

- natural environment
- context retained
 (observe २३३३ २३३ pattern change ३ २३३ २३३)
- longitudinal studies possible.

disadv : distraction, noise

Evaluating implementations :

obj/sub that will be evaluated ↔ artefact.

experimental factors :

subjects

variables

hypotheses → २ ३१ independent variable - २३३३

smoking ↔ programming

pos / neg / no relation.

lab works

23 research
 phd, master,
 field study
 ११११ ११११
 २२१



data / measurable $\rightarrow$ dependent var.
(skill / line of codes written)

smoke $\rightarrow$ independent

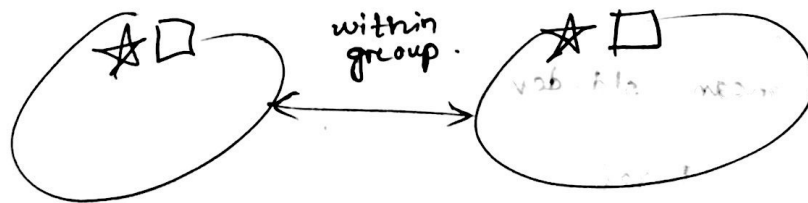
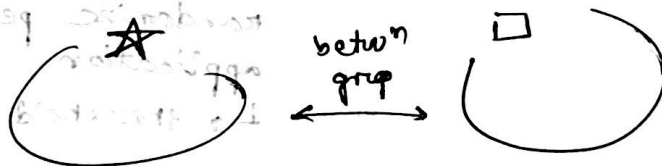
A & B has relation $\rightarrow$ pos

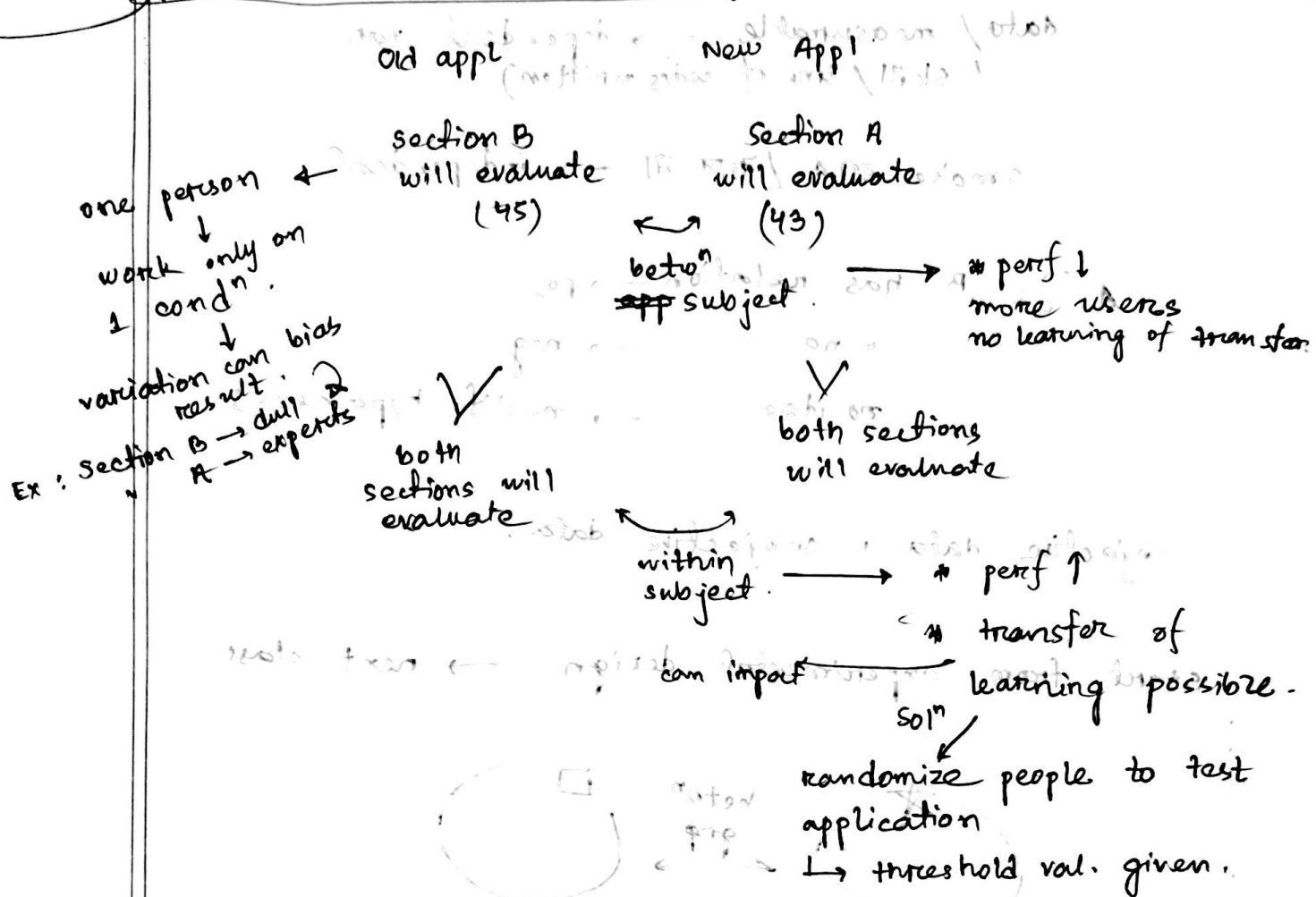
"no" $\rightarrow$ neg

no idea $\rightarrow$ nullify hypotheses.

objective data + subjective data.

start from experimental design $\rightarrow$ next class.



ImportantExperimental DesignAnalysis of Data :

mean, std. dev.

T val
p "

experimental studies on groups :

Safety
Societal Impact

Acceptable risk,
Accountability of
uncertainty

compliance
confidentiality
Conflict of Interest

Data integrity
Envi. health
Fairness
Honesty/Dishonesty

Short note

Think Aloud :

perform + describe (करना + बताना)

cooperative evaluation

evaluating system as a group

करना 2 side skip

Post-task walk-throughs

per task complete → tabulation

Post test → गुणा
करके complete करके
test → then decision

Interviews :

intellectual questions + qualitative data → takes time to transcribe

Questionnaires :

quick

subjective + qualitative data

↳ data generalize easy

but, it's difficult to prepose questions

less flexible → as answers fixed questions

+ semi-structured / unstructured
→ question ready
(novice people)
(prompt)
→ master + no need to create question

Eye tracking
physiological measure

} for behavior pattern

Imp

→ choosing eval method

⑧

Emergency app → route show करना

↳ evaluate this

Scenario based Q.

task usability

(1) method use to evaluate

(2) method that participant was.

for consideration.

if expe → betwⁿ group within

if heuristic → steps ✓

→ justify why it was chosen.

(see the prev. year questions & solve them)

subject → task better

objective → task perf better ?